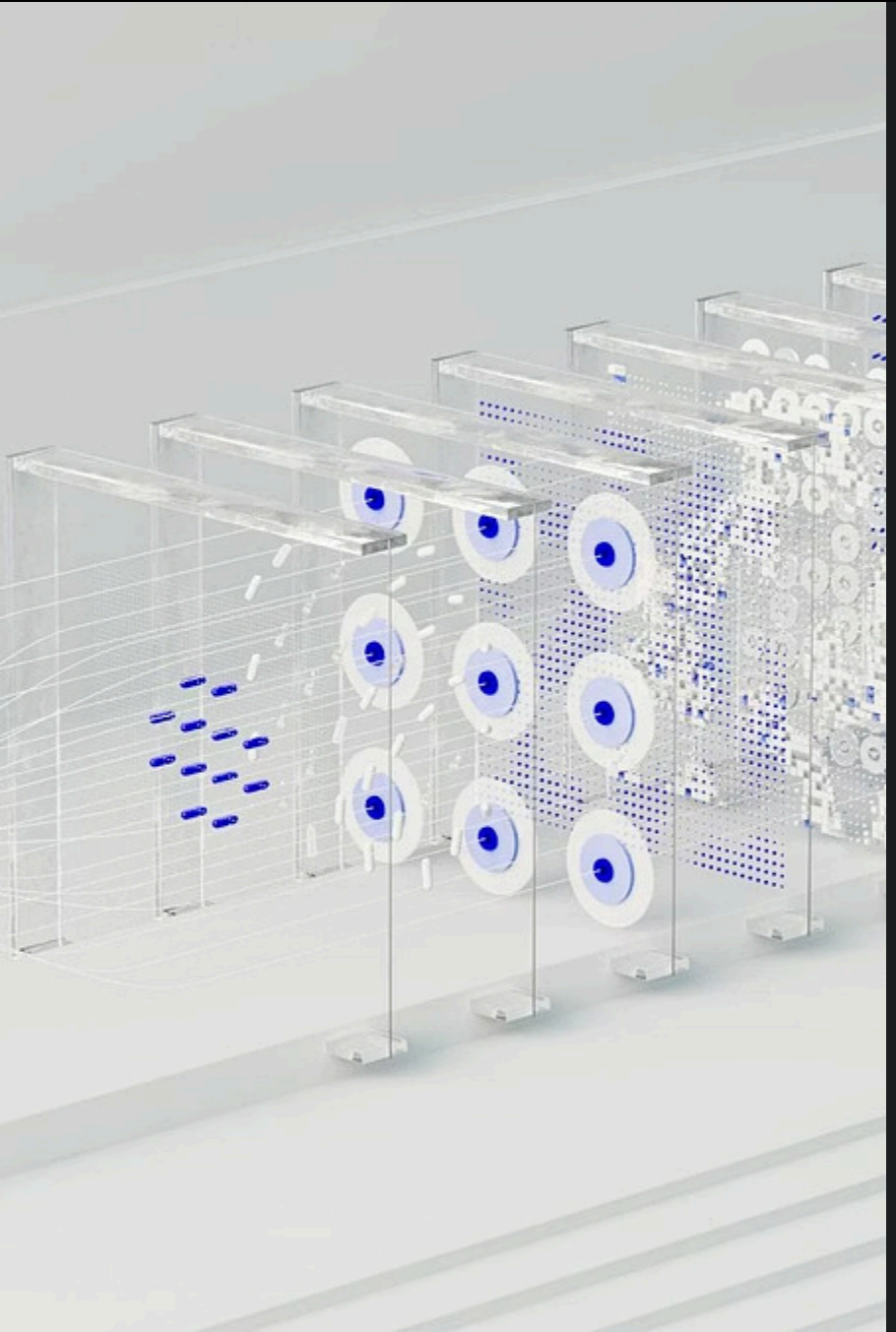

AI SAFETY & EVALUATION

Gen AI Safety & Risk Management Frameworks

A landscape overview of emerging standards, benchmarks, and evaluation strategies shaping responsible AI deployment

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CHAPTER 1

The Global Push Toward Standardized AI Safety

Industry, academia, and government are converging on shared frameworks to evaluate and manage generative AI risks — making safety an iterative, cross-sector discipline.

MLCommons

AI Safety Benchmark v0.5 — 7 hazard categories for chat-tuned LLMs

NIST

Generative AI Profile — Govern, Map, Measure, Manage framework

CETaS & Google

Sociotechnical evaluations & Critical Capability Levels (CCLs)

OpenAI

Structured scorecards & adversarial red teaming before deployment

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CHAPTER 2

Key Risk Categories Under Evaluation

1

Violent Crimes & Hate Speech

Content that incites violence or targets protected groups

2

CBRN Threats

Synthesis of Chemical, Biological, Radiological & Nuclear information

3

Confabulation

High-confidence generation of false or fabricated content

4

Human Uplift

AI providing malicious actors more capability than existing tools

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How the Frameworks Fit Together

Safety is not a single checkpoint — it is a continuous loop of identification, governance, evaluation, and adversarial testing across the entire AI lifecycle.

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Critical Questions Facing the Field

- **Benchmark Saturation**
  Models score near-perfect, making current tests obsolete as a progress measure
- **Language & Modality Gaps**
  Most benchmarks cover English text only — multilingual and multimodal coverage remains limited
- **Partial Unsafe Responses**
  If any part of a response is unsafe, the entire interaction is labeled unsafe

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Response Playbook: When a CCL Is Reached

Detect

Capability threshold identified through evaluation

Secure

High-level security mitigations — weight protection, access controls

Mitigate

Deployment safeguards — safety fine-tuning, monitoring, red teaming

Google's Frontier Safety Framework activates this protocol any time a model approaches severe-harm thresholds in cyber, bio, or autonomous capabilities.

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RECOMMENDATIONS

Actionable Recommendations

1 Adopt NIST's Govern-Map-Measure-Manage

Embed the AI RMF Generative AI Profile into your organizational risk lifecycle today

2 Implement Continuous Red Teaming

Move beyond one-time audits — establish recurring adversarial testing cadences tied to model updates

3 Plan for Benchmark Evolution

Track multilingual, multimodal, and sociotechnical evaluation advances to avoid saturation blind spots

4 Define Internal CCL Thresholds

Establish your own critical capability levels with pre-approved response playbooks before capabilities emerge

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REFERENCES & RESOURCES

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